Page 4 Practical Report

Orchestration Layer Kya Hai? | AI Coding Agents: Intern se Confident

Objective

Implement a small, testable JavaScript demonstration of an orchestration layer that coordinates a request, checks tool permissions, invokes an authorized tool, observes the result, and verifies acceptance criteria.

Implementation summary

Component	Implemented behavior
Request and criteria	Accepts a task goal and explicit acceptance checks.
Proposed action	Uses a deterministic demonstration action to select the local **filterTasks** tool.
Permissions	Blocks tools that are not included in the allowed-tools list.
Execution and observation	Runs the authorized local function and records the observed result in a trace.
Completion decision	Returns a **completed**, **incomplete**, or **blocked** outcome.

Orchestration flow

Human request → **Orchestrator** → **Proposed action** → **Permission check** → **Authorized tool** → **Observed result** → **Acceptance check** → **Handoff state**

The demonstration reflects the handbook's key distinction: a prompt can guide behavior, but application logic coordinates execution, enforces permissions, observes task state, and determines whether completion criteria are met.

Verification results

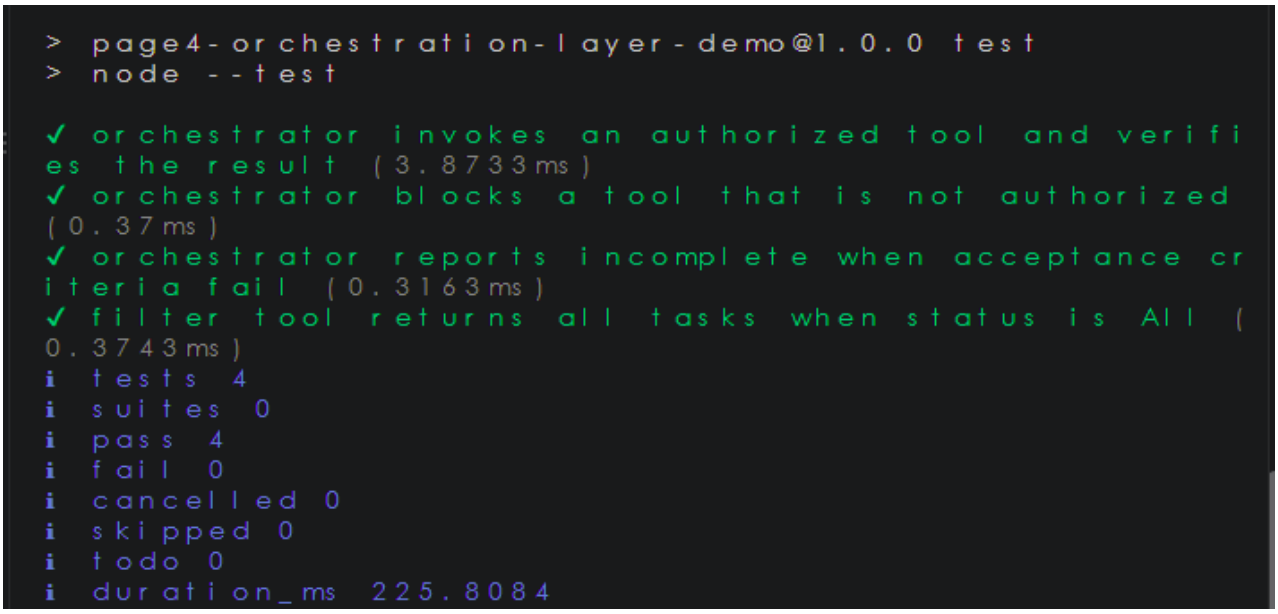
Tests	Passed	Failed	Result
4	4	0	**PASS**

Automated test coverage

Test case	Verified behavior
Authorized tool	The tool runs and the observed result satisfies the acceptance criteria.
Unauthorized tool	The proposed tool is blocked by the permission check.
Incomplete criteria	The task is marked incomplete when the criteria are not satisfied.
All status filter	The filter returns all local fake tasks when status is set to All.

Evidence — local test run

The screenshot below shows the test command run on the project and the result: 4 tests passed, 0 failed.



Scope and limitations

This is an educational simulation. It does not connect to a real LLM, repository, database, network, or external shell tool.

Conclusion

The Page 4 orchestration demo has been implemented and verified with four passing tests. The key learning outcome is that orchestration coordinates tool execution, enforces permissions, observes results, and checks completion criteria; a prompt alone does not provide these application capabilities.